

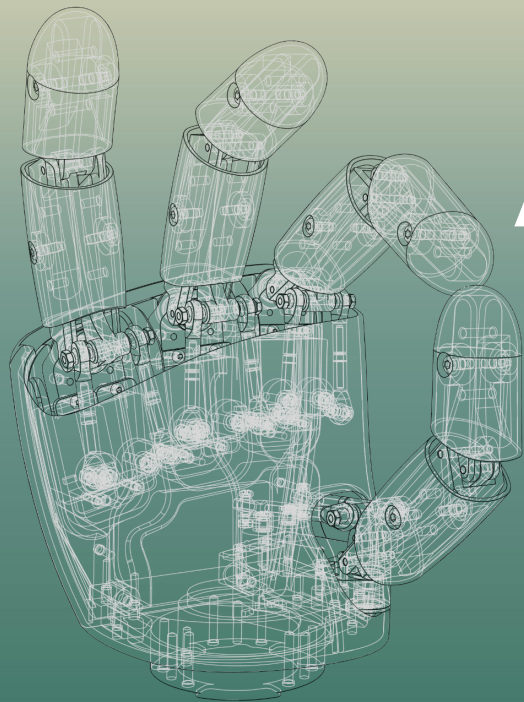
GOSIM

GOSIM Paris 2026

GOSIM Paris 2026

May 5-6, 2026 Station F, Paris

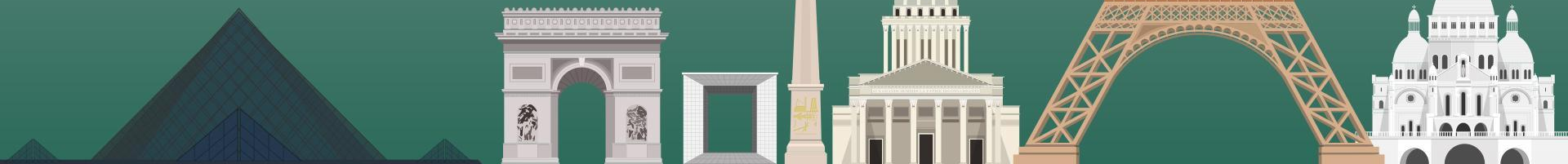




Amazing Hand

Open source robotic hand
tech report

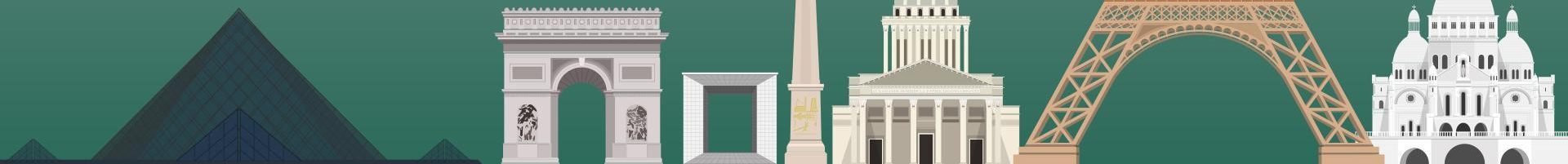
Jeremy Laville



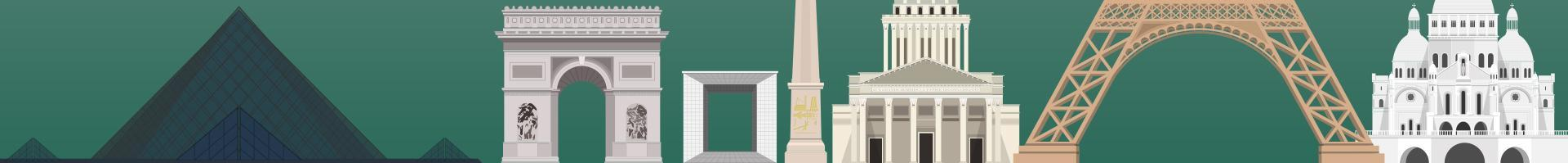
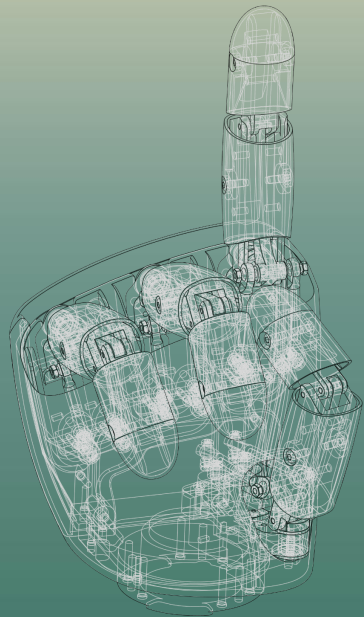
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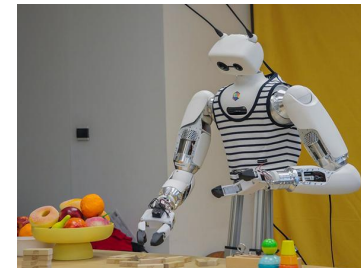


Introduction

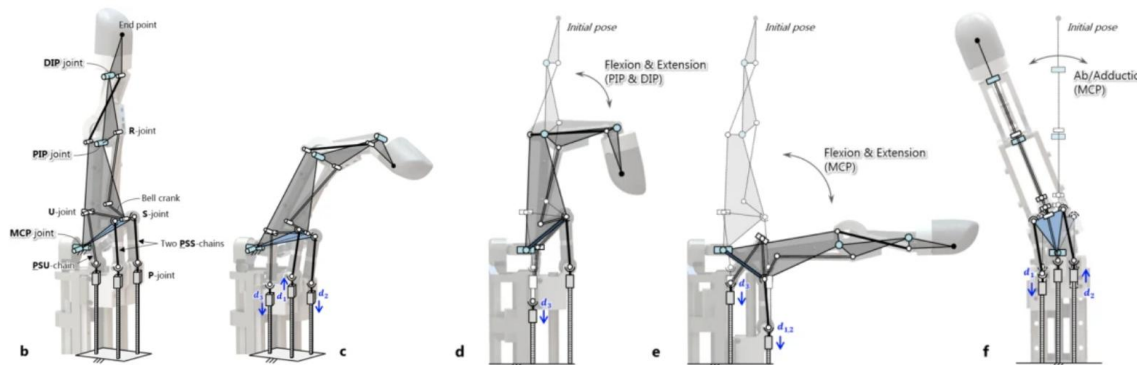


Beginning of 2025 : Pollen's internal hackathon

Goal : Design a low cost robotic hand with expressive and grasping capabilities with all actuators embedded inside the hand itself, suitable for Reachy2 Pollen's robot



Inspiration : ILDA Hand (Integrated Linkage-driven Dexterous Humanoid) from South Korean Lab (Iir Lab)



=> Impressive capabilities, with smart parallel mechanism

But : Expensive tiny linear actuators
Custom mechanical parts (CNC machined)
Custom Pcb's to control actuators & sensors



Key features :

Low cost

Grasping

Expressive

Standalone

Easy to build

Technical choices

=> Hobby grade servomotors + fully 3D printable

=> Flexion / Extension + Opposable thumb

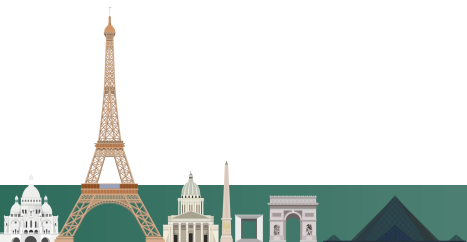
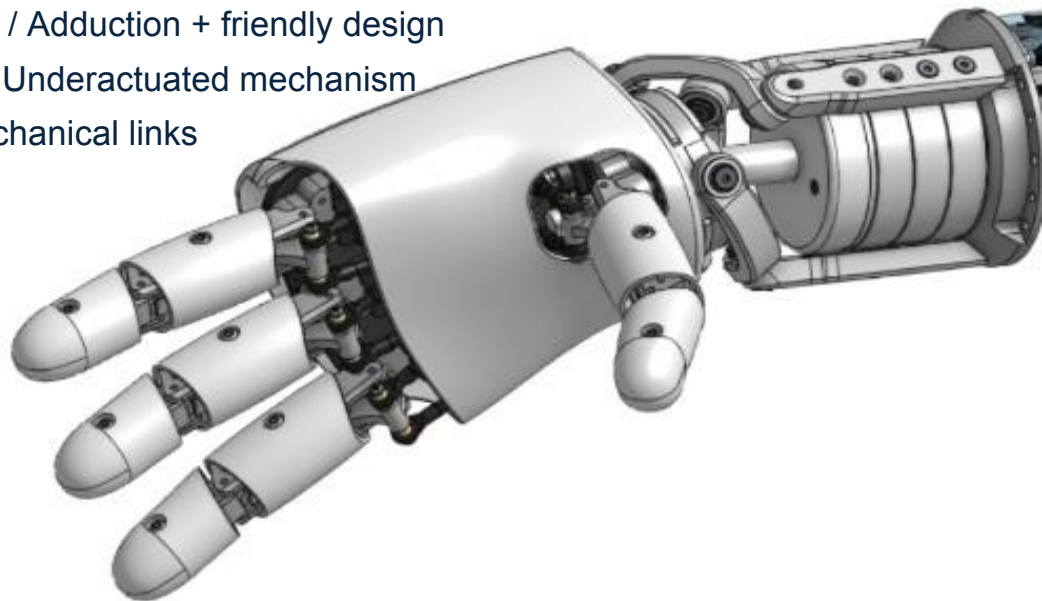
=> Abduction / Adduction + friendly design

=> Parallel & Underactuated mechanism

=> No tendons, direct mechanical links



8 dofs robotic hand
fully open source
< 400 g weight
< 200 \$ cost



July 2025 : Open source release on github : <https://github.com/pollen-robotics/AmazingHand>
& Article on Hugging Face blog : <https://huggingface.co/blog/pollen-robotics/amazing-hand>

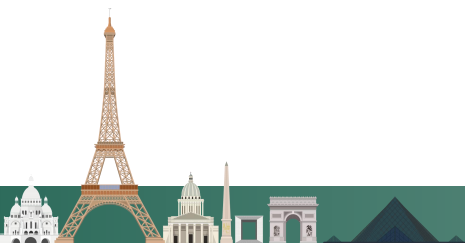
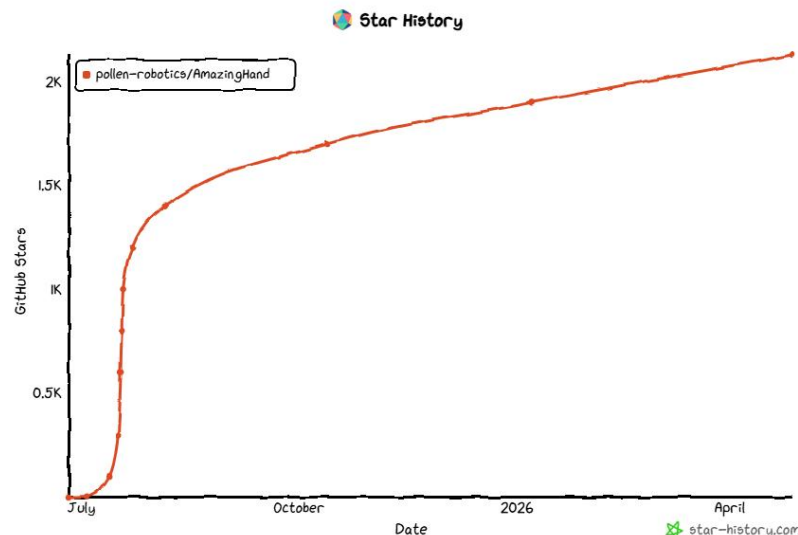
Project is licensed under Apache 2.0 & Mechanical design is licensed under a Creative Commons Attribution 4.0

BOM, CAD , Printing guide, Assembly guide, Basic demo (arduino / python), Advanced demo available

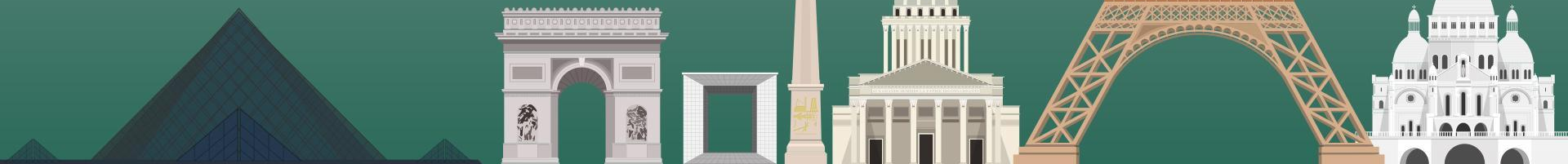
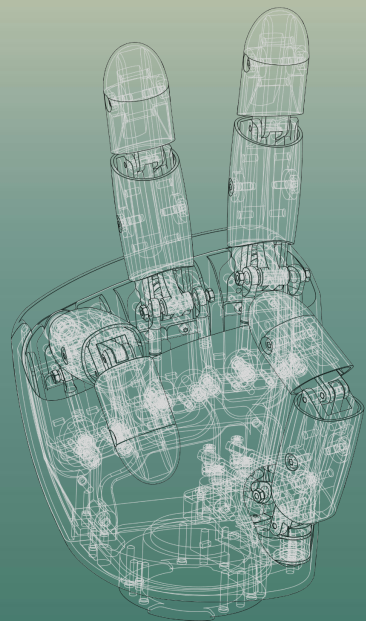
Since the release :

- More than 2K stars reached, 217 forks
- Strong community around the project
- Kits available under 150 \$
(WowRobo / Seeed Studio / Aliexpress...)

Few weeks ago : New version “Enhanced” released



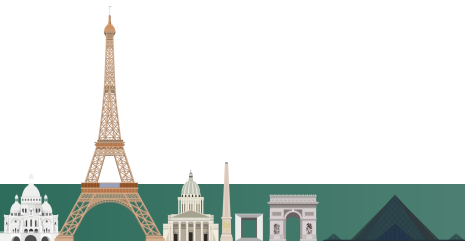
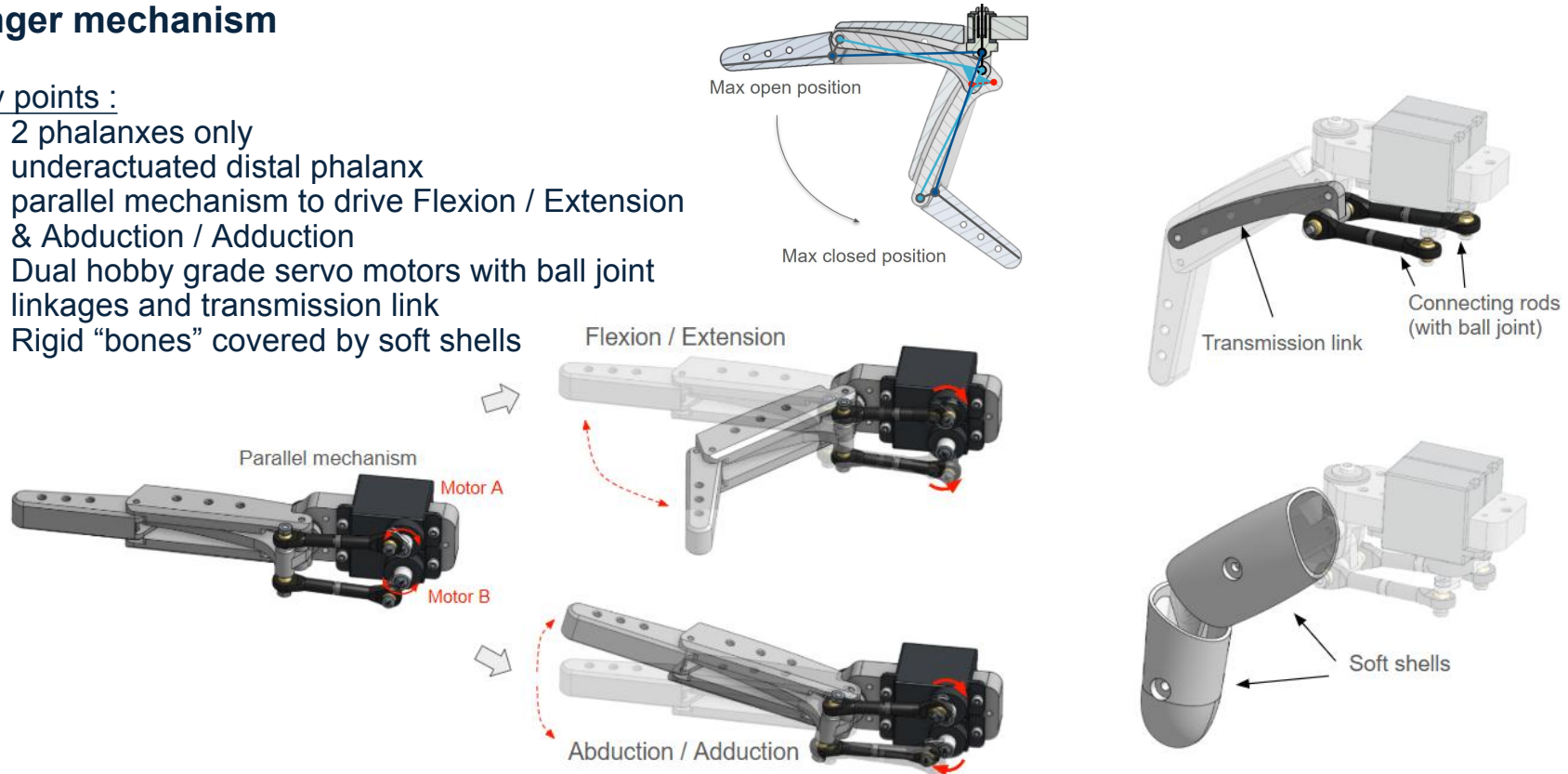
Technical choices & Result



Finger mechanism

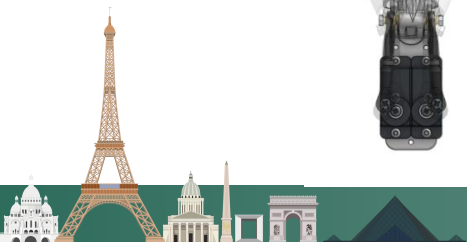
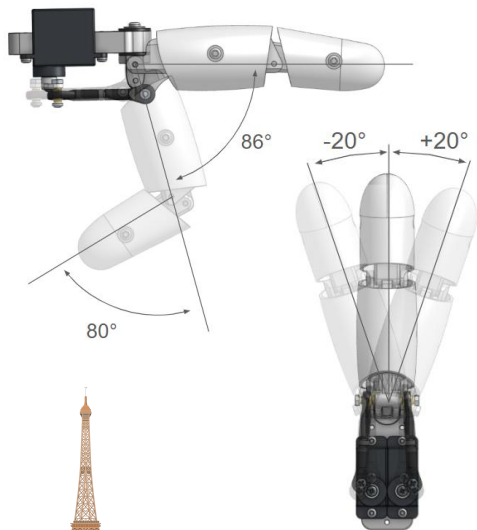
Key points :

- 2 phalanxes only
- underactuated distal phalanx
- parallel mechanism to drive Flexion / Extension & Abduction / Adduction
- Dual hobby grade servo motors with ball joint linkages and transmission link
- Rigid “bones” covered by soft shells



Finger mechanism

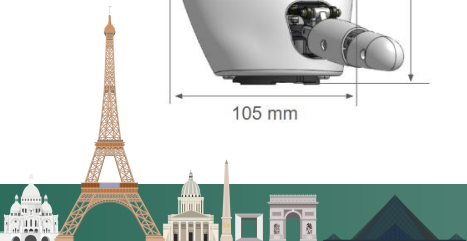
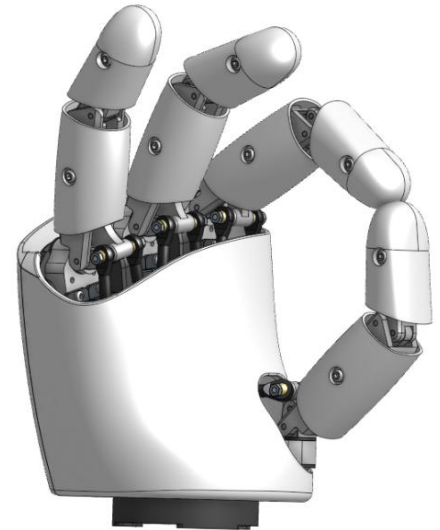
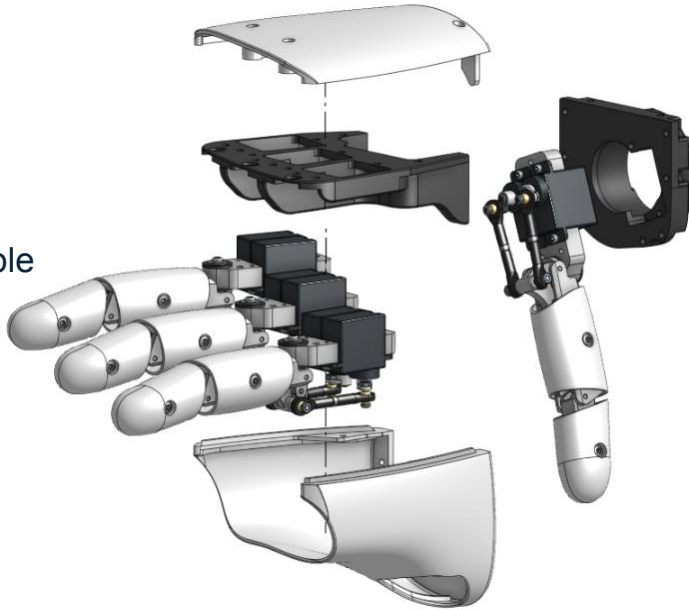
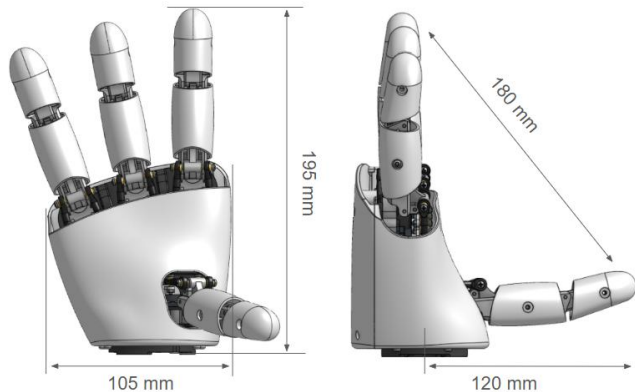
Main challenge was to balance between :
range of flexion / extension
+ range of abduction / adduction
versus servo horn length
by minimizing size (specially width)



Hand design

Key points :

- Standalone finger unit x4
- Structural part + Wrist interface
- Human size (Adult)
- Index and thumb fingers are opposable
- Covering with soft palm



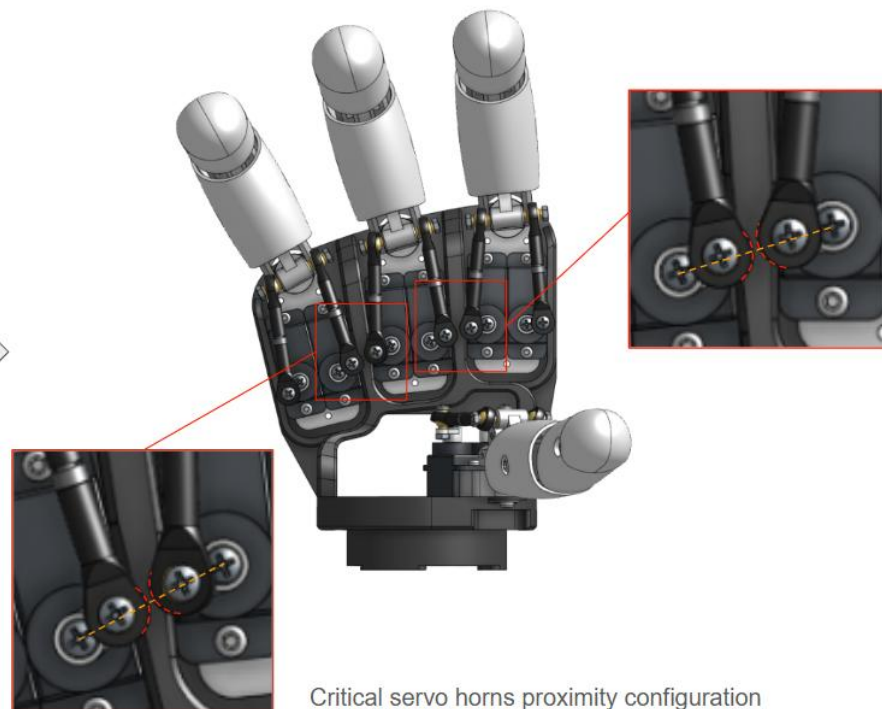
Hand design

First challenge was to balance between :

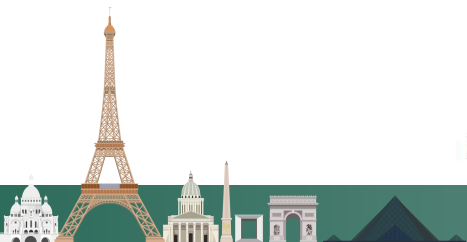
Number of fingers
versus Keeping human size



Nominal positions of all fingers & thumb



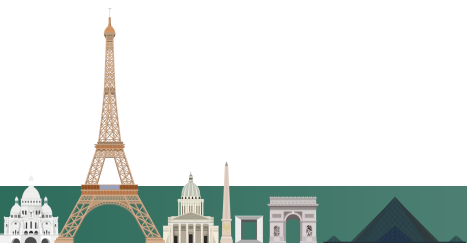
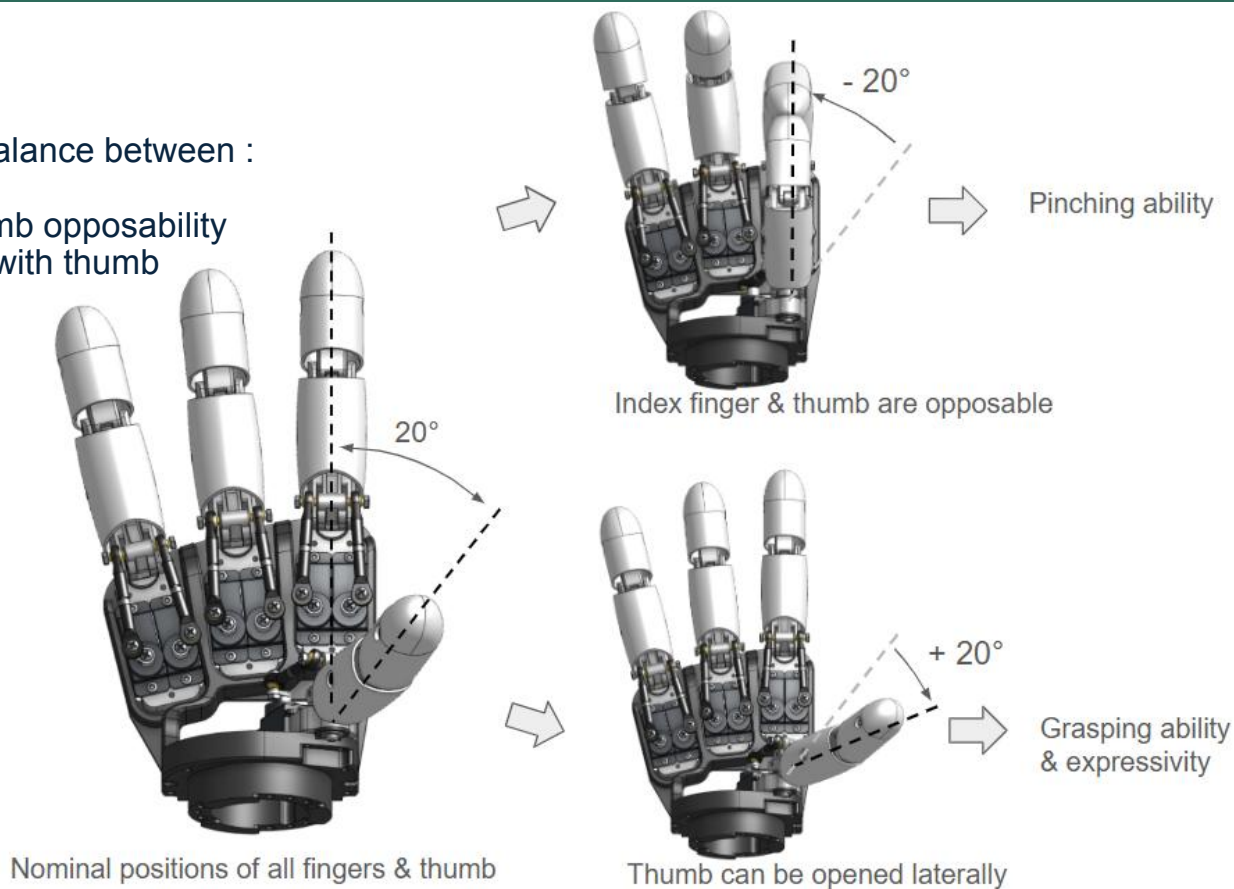
Critical servo horns proximity configuration



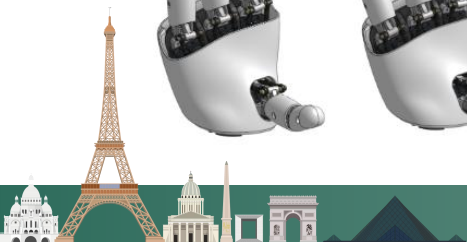
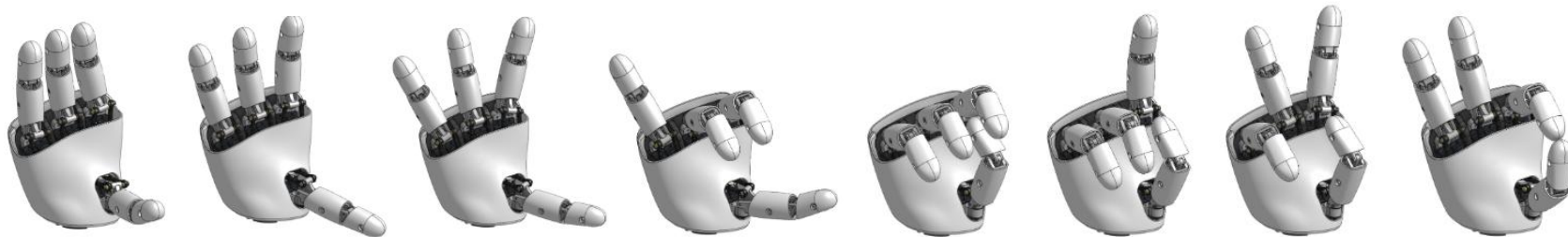
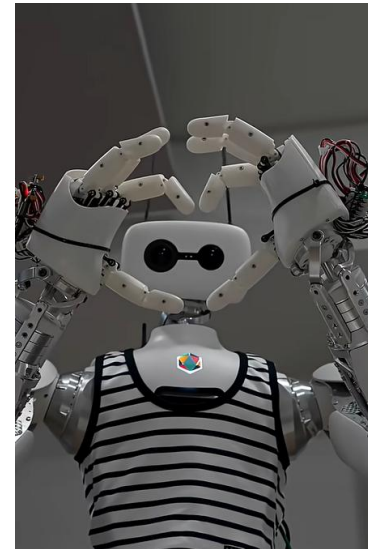
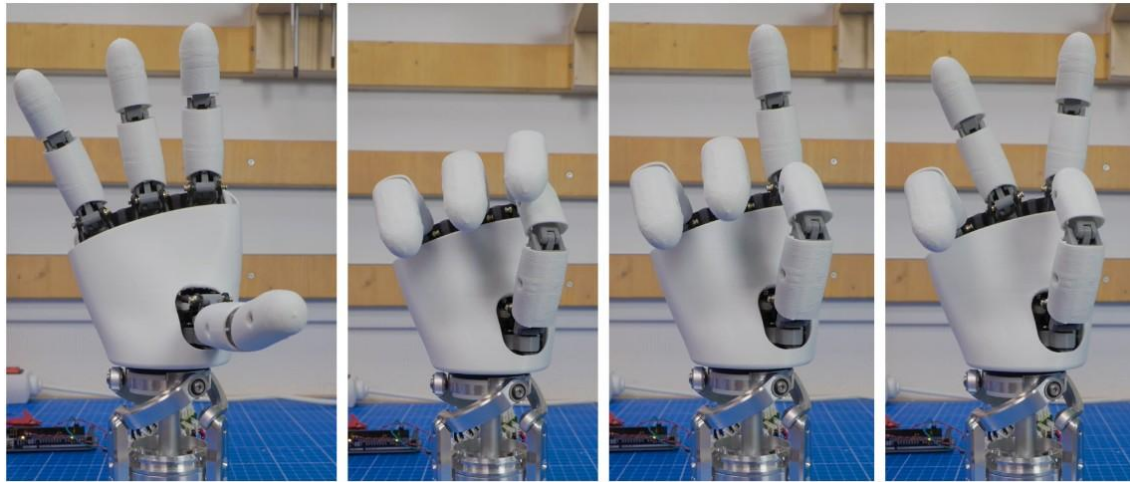
Hand design

Second challenge was to balance between :

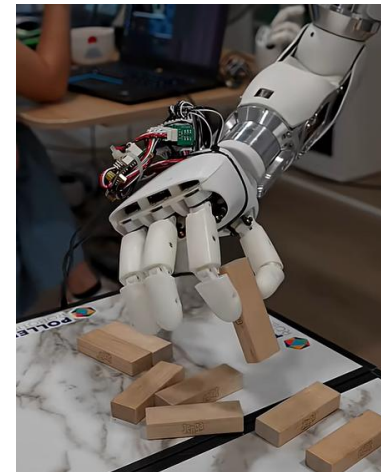
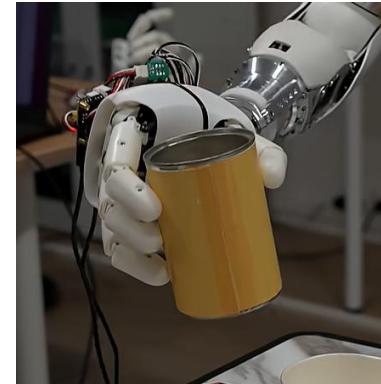
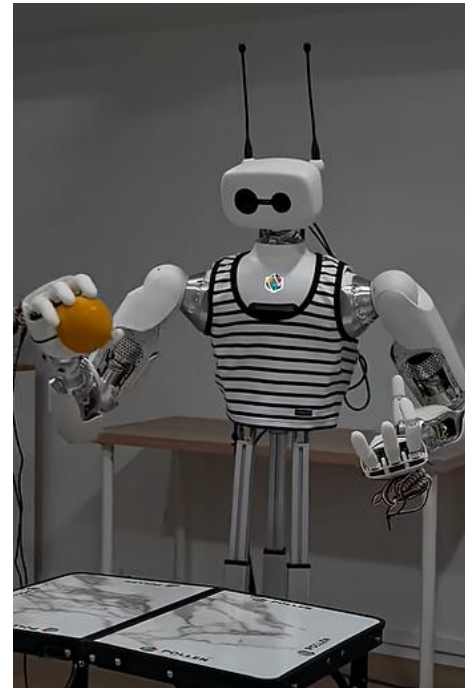
Index finger & Thumb opposability
versus Expressivity ability with thumb



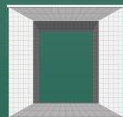
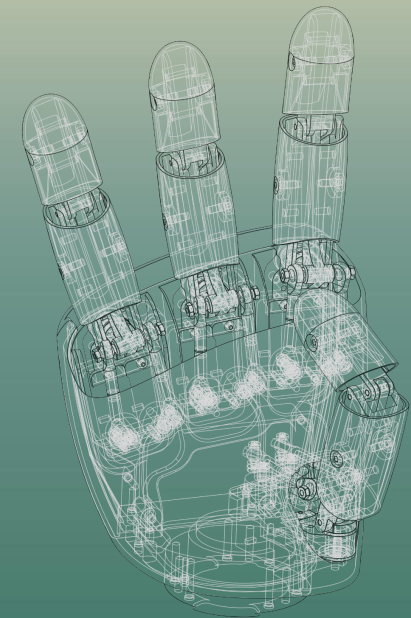
Expressivity



Grasping

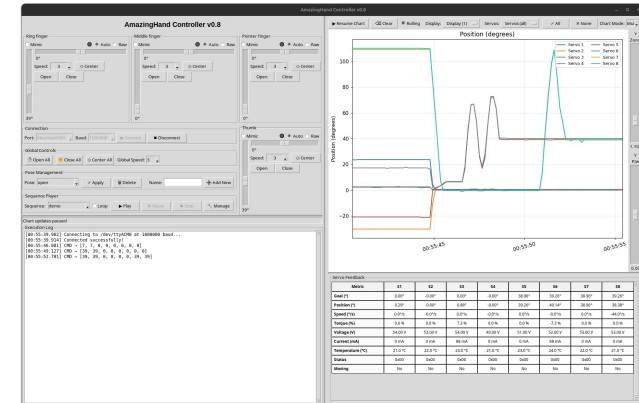
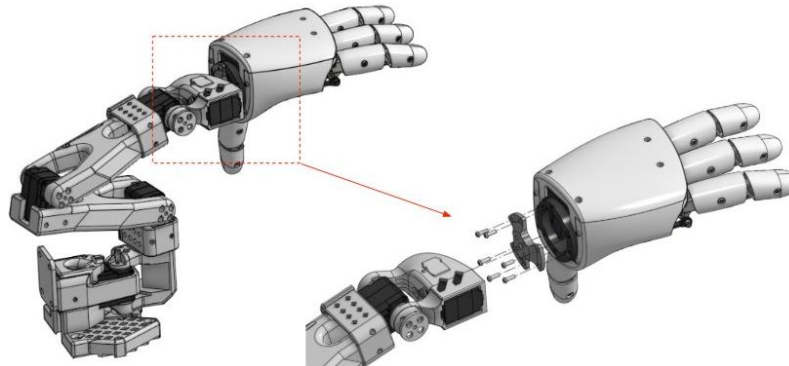


Community updates & Feedback



Several community updates :

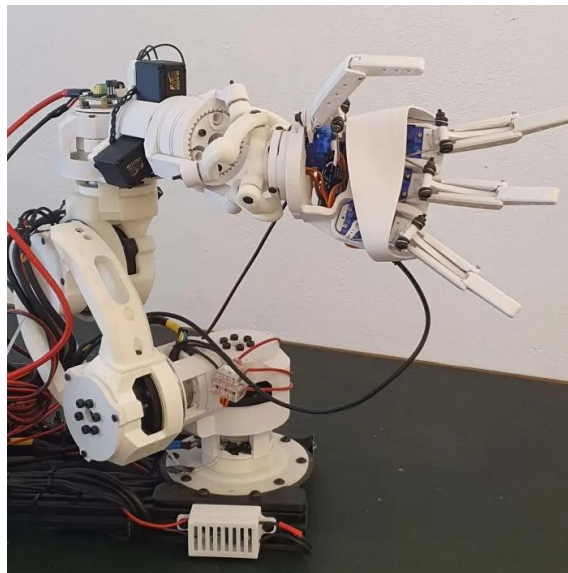
- Base for Amazing Hand
- Torque control with 9g servos
- Interface part for SO ARM
- Python GUI and CLI tools



Several robotic arms :



SO ARM projects



[Mickey666Maus](#)

open source 7 dofs arm



Aifitlab

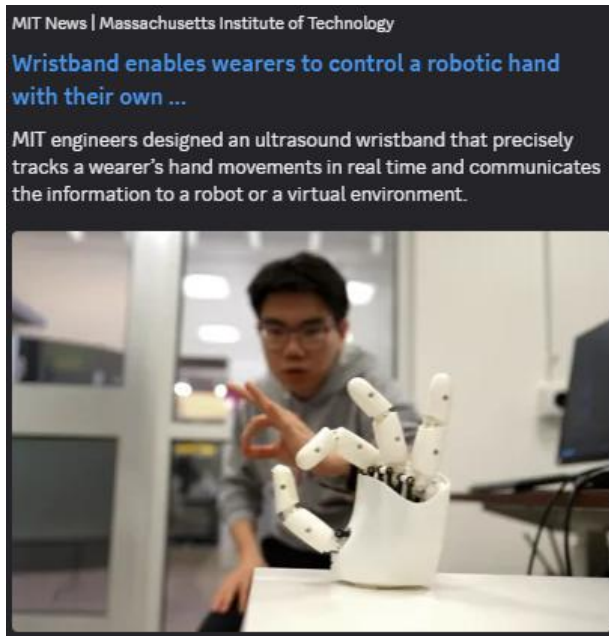
6 dofs commercial grade arm



Several projects :



BruBotics (Belgium)
Tactile sensor integration

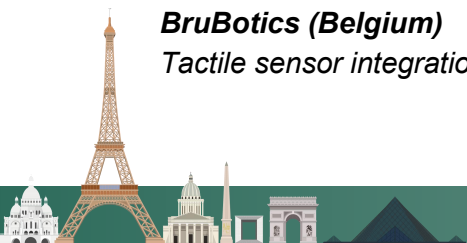


MIT (USA)
Ultrasound wristband control

The Thousand-Hand Robot, an emotional interaction robot jointly launched by Tsinghua University's Future Laboratory and Qingmin Perception, is driven by the Qwen large model and works in tandem with intelligent robotic hands to enable voice-based Q&A communication and interactive rock-paper-scissors gameplay.



Tsinghu University's future lab (China)
Interactive communication



Several feedback :

Cons

“Truly amazing”
in fingers to make precise tasks”
“Perfect for educational topics”

Pro

“too much backlash

“no backdrivability”

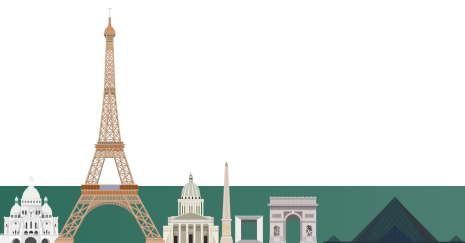
=> **Enhanced** version released few weeks ago, with better motors which offer :

- Full backdrivability
- More torque / speed
- More reliable / less backlash
- Less noise

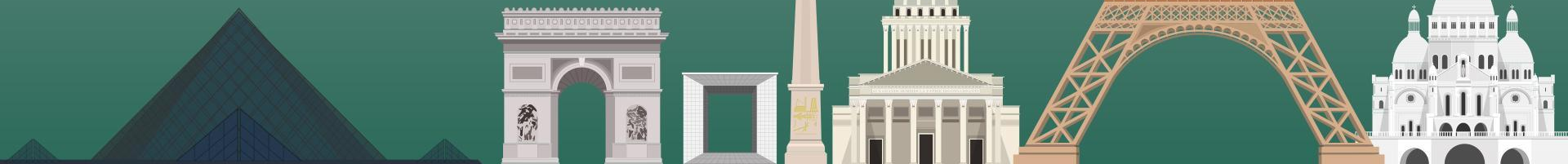
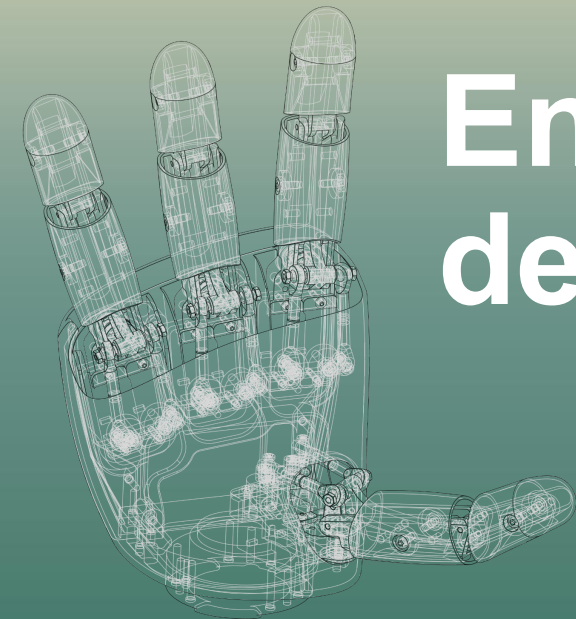
Price increased too, but still remain reasonable (<350



Specification	SCS0009	STS3032
Dimensions	23.2 × 12.1 × 25.5 mm	23.2 × 12.1 × 28.5 mm
Weight	~21 g	~20 g
Max torque	2.3 kg·cm	4.5 kg·cm
Rotation range	300°	360° / multi-turn
Resolution	10-bit (1024)	12-bit (4096)
Case	Plastic	Aluminum
Gear material	Mixed	Full metal
Motor	Standard DC	Coreless
Speed	0.07 s/60°	0.09 s/60°



Enhanced version demo

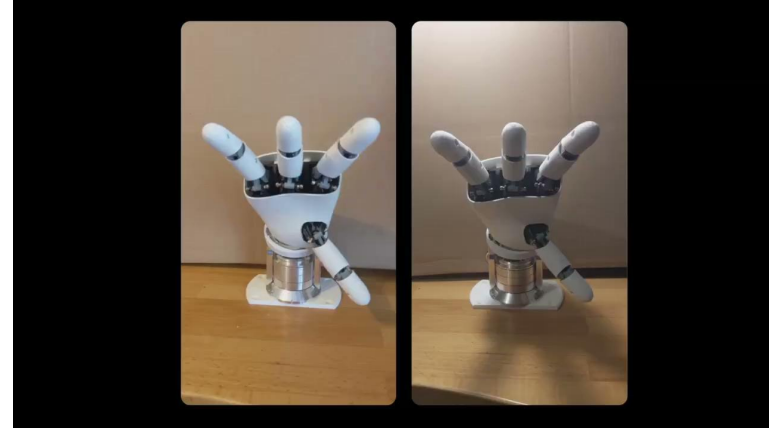
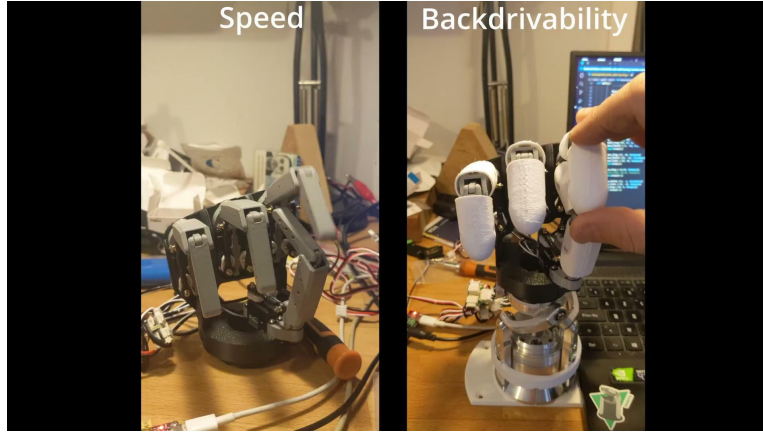


03- Enhanced version demo

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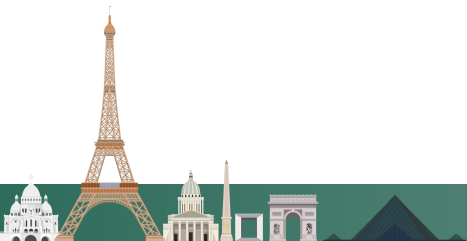
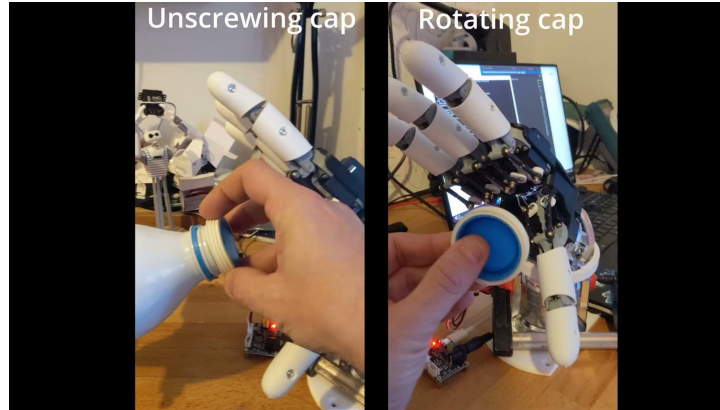
Speed

Backdrivability



Unscrewing cap

Rotating cap



Thanks!

